ACEBOTT

Tutorial for Using the Motion-Sensing Glove

Preface

Our Company

ACEBOTT STEM Education Tech Co.,Ltd

Company History: Founded in 2013, located in China's Silicon Valley—Shenzhen. We have assembled a team of 150 members, including professionals in R&D, production, sales, and logistics. Our goal is to provide customers with outstanding STEM education products and services. We collaborate with STEM education experts and business partners worldwide to deliver premium STEM education kits, while also offering OEM services, including product packaging and PCB logo customization.

Tutorial

This is a hands-on course designed for beginners, aimed at guiding students into the world of programming, electronics, and robotics through the use of a motion-sensing glove. In this course, students will learn the theoretical knowledge and practical applications of motion-sensing glove control, and use the glove to control QD series smart cars, bipedal robots, and more.

With this kit, you can:

1.Understand the structure and functions of the motion-sensing glove.

2.Learn the basic principles of Bluetooth signal transmission and master the application of Bluetooth communication technology.

3.Master the method of controlling robot movement using the motion-sensing glove.

4.Use the ACEBOTT kit to complete projects such as controlling QD series cars and bipedal robots with the motion-sensing glove, enhancing your maker skills.

After-Sales Service

ACEBOTT is a dynamic and rapidly growing STEM education technology company dedicated to providing excellent products and high-quality services to meet your expectations. We value your feedback and encourage you to send any opinions or suggestions to **support@acebott.com**.Our experienced engineering team is committed to quickly resolving any issues or questions you may encounter while using our products. During weekdays, we guarantee a response within 24 hours.

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YouTube QR Code

Contents

Lesson 1 Motion-Sensing Glove

With the development of technology, robots are playing an increasingly important role in many fields. However, the operation of traditional robots usually relies on complex programming or bulky control devices, which not only limits the flexibility of operation, but also increases the cost of learning and use. With the development of technology, Motion-Sensing Glove, as a cutting-edge tool for human-computer interaction, are changing this situation. It captures the natural movements of the hands and converts them into control instructions for the robot, allowing the operator to interact with the robot in a more intuitive and efficient way.

Next, we will understand the working principle of the Motion-Sensing Glove and learn how to use it.

1. Introduction to Motion-Sensing Glove

1.1 Function of Motion-Sensing Glove

ACEBOTT Motion-Sensing Glove are wearable devices that can capture hand gestures and postures and convert these data into digital signals for interaction.

1.2 Application Scenarios of Motion-Sensing Glove

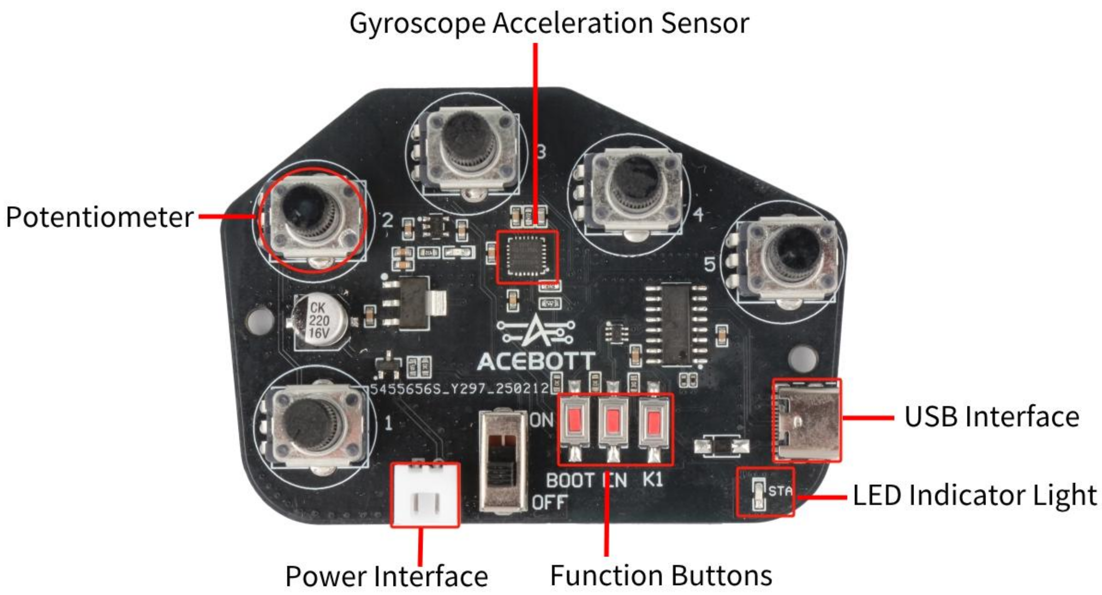
(1) Remote Control and Robot Control

Through the Motion-Sensing Glove, manipulators or robots can be controlled to achieve precise remote control, which can be applied in surgical operations, industrial production, rescue and other scenarios.

(2) Virtual Reality (VR) and Augmented Reality (AR)

Allow users to interact more naturally in VR/AR environments, such as grabbing virtual items, controlling the movement of virtual characters, etc.

2. Motion-Sensing Glove Controller Board



ESP32 Controller: As the core processor, it is responsible for data collection and processing, and has Wi-Fi and Bluetooth functions. (It is located on the back of this chip, near the switch)

Potentiometer: used to detect the degree of finger bending and output resistance value by changing the rotation angle.

Gyroscope Acceleration Sensor: used to detect hand posture and movement.

Function Keys: There are three function keys, namely BOOT key, RST key and K1 key.

BOOT Key: When the controller is to burn firmware, you need to press this key to put the controller in download mode and then burn the firmware.

RST Key: Reset key, which allows the controller to restart the program.

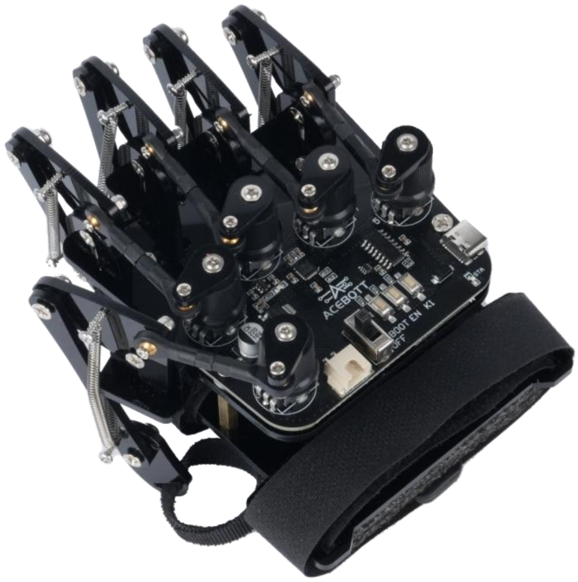
K1 Key: Function key, users can customize its function.

LED Indicator Light: used to display the connection status between the gloves and the control device.

USB Interface: used for debugging and downloading programs on the PC.

Power Interface: used to connect external power supply, power range is 3V-12V.

3. Motion-Sensing Glove Structure

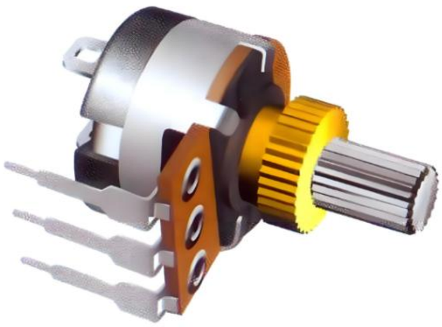


ACEBOTT's Motion-Sensing Glove consist of five connecting rod components and a base plate. Each connecting rod is connected to a specific finger and a potentiometer. When the finger bends, the connecting rod drives the potentiometer knob to move, thereby changing its resistance value. The controller can accurately capture the finger posture information by analyzing the change in resistance value.

The control panel is mounted on the bottom plate and fixed to the back of the user's hand. When the user's hand moves, the gyroscope on the control panel can detect the direction and speed of the palm movement.

4. Potentiometer

Potentiometer is a variable resistor, commonly used to adjust the voltage or current in the circuit. It consists of a resistor and a movable slider. By rotating or sliding the slider, the output voltage can be changed to control the working state of the device. Common potentiometers include knob potentiometers, slide potentiometers, and digital potentiometers. The device uses a knob potentiometer.



5. Gyro Acceleration Sensor

The gyroscope acceleration sensor model used in this device is MPU6050, which is a 6-axis inertial measurement unit (IMU) that integrates a 3-axis accelerometer and a 3-axis gyroscope. It can measure acceleration and angular velocity and is widely used in robots, drones, smart cars, somatosensory control and other fields.

MPU6050 main functions:

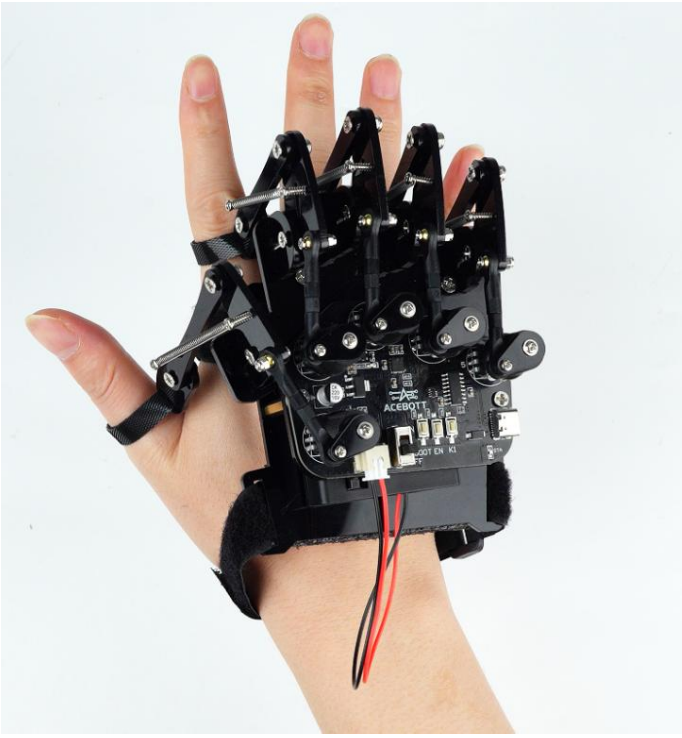
Sensor	Effect	Unit
Accelerometer	Measure linear acceleration (front and back, left and right, up and down)	g

Gyroscope	Measures angular velocity (speed of rotation around the X, Y, and Z axes)	°/s
Temperature Sensor	Measuring chip temperature	°C

The accelerometer can be used to detect tilt angle, vibration, and free fall, while the gyroscope is mainly used to measure the rotation speed of an object. When the two are combined, the attitude angle of the device can be calculated.

6. How to Wear the Motion-Sensing Glove

Open all the buckles, first fix the buckle on the connecting rod at the last knuckle of the five fingers, then fix the palm buckle and wrist buckle at the palm and wrist respectively. The final fixing effect is as follows.



Click here to view the glove wearing instructions document.

Note: To watch the glove wearing video, please click the link below and select the corresponding video to view.

https://www.youtube.com/playlist?list=PLkW5fEtHNu6JIQVJpGaInKdkuk6Er4dIN

7. Motion-Sensing Glove Instructions

Each gesture and posture corresponds to a command signal. When the motion sensing glove detects a gesture, it will broadcast a corresponding command to other devices.

7.1 Motion-Sensing Glove Command Library

The gesture command library of this device is as follows:

Hand Gesture Pictures	Gesture Description	Command Name
/	/	command0
	Extend your index finger, palm down, and tilt upward	command1
	Extend your index finger, palm down, and tilt it downward	command2

	Extend your thumb and index finger, palm down, lean to the right	command3
	Extend thumb, index finger, palm down, left leaning	command4
	Extend your index and middle fingers, palm down, and tilt upward.	command5
	Extend your index and middle fingers, palm down, and tilt your hand downward.	command6
	Extend your index finger, middle finger, ring finger, and pinky finger, palm facing down, leaning left	command7
	Extend your index finger, middle finger, ring finger, and little finger, palm down, and lean to the right	command8
	Extend your index and middle fingers, palm facing right, left	command9
	Extend your index and middle fingers, with your palm facing left and right	command10
	Open your palms, palms down, lift them up	command11
	Open your palms, palms down, press down	command12
	Make a fist and rotate your wrist counterclockwise, with your palm facing right.	command13
	Make a fist and rotate your wrist clockwise, with your palm facing to the left.	command14

	Open your hand, palm down, palm tilted to the left	command15
	Open your hand, palm down, palm tilted to the right	command16
	Extend your thumb and index finger, with your palm facing left and up.	command17

7.2 Motion-Sensing Glove Command Test

Testing the gesture detection function of the motion-sensing glove via Arduino serial port. Click here to obtain the command test program.

The program is as follows:

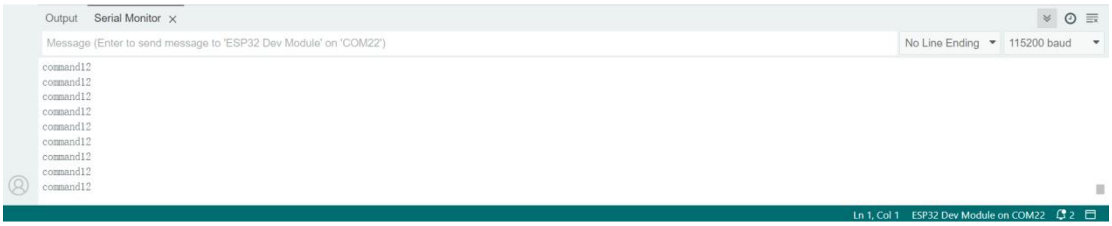
```
#include <ACB_Gloves.h>
#include <ACB_BLEDevice.h>

ACB_Gloves gloves;       // gloves object
ACB_BLEDevice ble;       // BLE object

// Setup function to initialize the server and characteristics
void setup() {
  Serial.begin(115200);   // Start the Serial communication for debugging
  gloves.init();
}

void loop() {
  gloves.get_data();      // GET DATA
  Serial.println(gloves.get_command());     // GET COMMAND
}
```

Please wear the motion-sensing glove and connect the glove's controller board to the computer using a USB cable. Then, upload the command test program to the glove's controller board.Next, open the Serial Monitor in the Arduino IDE to observe the output command values.Finally, refer to the command library provided above to verify whether the gestures match the corresponding commands.



Lesson 2 Motion-Sensing Glove Communication

In order to achieve efficient communication between the Motion-Sensing Glove and the robot and ensure that the hand movements can be converted into the robot's control instructions in real time and accurately, communication technology plays a vital role. This device will use the Bluetooth function of ESP32 to communicate with the robot. Next, we will explore how the Motion-Sensing Glove communicates with other devices and controls them.

1. Bluetooth Communication Principle

Bluetooth is a short-range wireless communication technology, mainly used for data transmission and communication between devices. Bluetooth technology is based on radio frequency (RF) signals, operates in the 2.4 GHz ISM (industrial, scientific and medical) frequency band, and uses frequency hopping spread spectrum (FHSS) technology to reduce interference and improve communication reliability.

Bluetooth communication technology is divided into classic Bluetooth and low-power Bluetooth (BLE) according to its functions and characteristics.

1.1 Classic Bluetooth

Classic Bluetooth is an early form of Bluetooth technology, mainly used in high-bandwidth, continuous transmission application scenarios, such as audio transmission, file transmission, etc. It has high transmission rate and high power consumption.

1.2 Bluetooth Low Energy

Bluetooth Low Energy is a technology introduced by Bluetooth 4.0, focusing on application scenarios with low power consumption and short transmission time, such as sensors, smart home devices and other IoT applications.

The Bluetooth communication of this device adopts low-power Bluetooth. The following will focus on the communication process of low-power Bluetooth.

2. Bluetooth Low Energy (BLE) Communication Process

Low-power Bluetooth devices communicate using a broadcast mechanism. The specific process is as follows:

2.1 Device Discovery and Connection

Broadcast: The slave device actively sends a broadcast packet to announce its existence.

Scan: The master device scans for nearby broadcasting devices.

Connection: The master device sends a connection request to establish a connection with the slave device.

2.2 Data Transmission

After the connection is established, the master device can read or write the characteristic value of the slave device.

The slave device can also actively send data to the master device through notification or indication.

2.3 Disconnect

When the communication is completed, the master or slave device can actively disconnect.

3. Motion-Sensing Glove Command Sending Program

The Motion-Sensing Glove transmits the detected gesture commands to the control device via Bluetooth communication.

Click here to get the instruction sending procedure.

Please put on the Motion-Sensing Glove and connect the

Motion-Sensing Glove controller board to the computer using a USB cable. After that, upload the command sending program to the Motion-Sensing Glove controller board.

Lesson 3 Motion-Sensing Glove to Control QD001 Movement

In the previous content, we have already mastered how the Motion-Sensing Glove detects gestures and how to send commands. So how does the robot receive these commands and then be controlled by the Motion-Sensing Glove?

Next, we will take the basic movements of the QD001 car controlled by the Motion-Sensing Glove as an example to decipher how the car is controlled by the Motion-Sensing Glove.

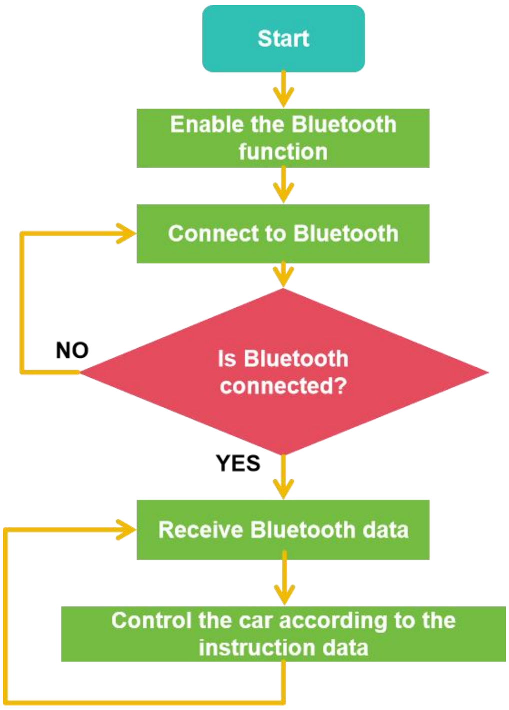
1. Motion-Sensing Glove Control the Car Logic

Hand Gesture Pictures	Gesture Description	Gloves Instructions	Car Action
	Open your palms, palms down, lift them up	command11	Go Ahead
	Open your palms, palms down, press down	command12	Back
	Make a fist and rotate your wrist counterclockwise, with your palm facing right.	command13	Rotate Left
	Make a fist and rotate your wrist clockwise, with your palm facing	command14	Rotate Right

	to the left.		
	Open your hand, palm down, palm tilted to the left	command15	Move Left
	Open your hand, palm down, palm tilted to the right	command16	Move right
/	/	command0	stop

2. Program for Controlling the Car with the Motion-Sensing Glove

Similarly, the car also needs to receive instructions from the Motion-Sensing Glove via Bluetooth. The program flow chart is as follows:



[Click here to get the car control program.]

The reference procedure is as follows:

```
#include <ACB_BLEDevice.h>
#include <vehicle.h>

#define SERVICE_UUID "4fafc201-1fb5-459e-8fcc-c5c9c331914b"    // Unique identifier for the service
#define CHARACTERISTIC_UUID "beb5483e-36e1-4688-b7f5-ea07361b26a8"    // Unique identifier for the characteristic
vehicle Acebott;   // Car object
ACB_BLEDevice ble(SERVICE_UUID,CHARACTERISTIC_UUID);    // Ble object

void setup() {
  Serial.begin(115200);  // Start the serial communication for debugging
  ble.Bluetooth_Connected("ESP32-BLE");    // bluetooth connected
  Acebott.Init();            // Initialize Acebott
  Acebott.Move(Stop, 0);  // Stop the Acebott exercise
}

void loop() {
  ble.ClientReceived();    // Received data
  if (!ble.connected) {
    Acebott.Move(Stop, 0);     // stop
  }
  // If connected and characteristic is writable, send data
  if (ble.connected && ble.pRemoteCharacteristic && ble.pRemoteCharacteristic->canWrite()) {
    delay(5);
    if (strcmp(ble.ClientData.c_str(), "command11") == 0) {
      Acebott.Move(Forward, 255);
    } else if (strcmp(ble.ClientData.c_str(), "command12") == 0) {
      Acebott.Move(Backward, 255);
    } else if (strcmp(ble.ClientData.c_str(), "command15") == 0) {
      Acebott.Move(Move_Left, 255);
    } else if (strcmp(ble.ClientData.c_str(), "command16") == 0) {
      Acebott.Move(Move_Right, 255);
    } else if (strcmp(ble.ClientData.c_str(), "command13") == 0) {
      Acebott.Move(Contrarotate, 255);
    } else if (strcmp(ble.ClientData.c_str(), "command14") == 0) {
      Acebott.Move(Clockwise, 255);
    } else if (strcmp(ble.ClientData.c_str(), "command0") == 0) {
      Acebott.Move(Stop, 0);
    }
    ble.ClientData = "";
  }
}
```

Turn on the power of the car, use the USB cable to connect the car's controller board and the computer, and upload the program to the car's controller board.

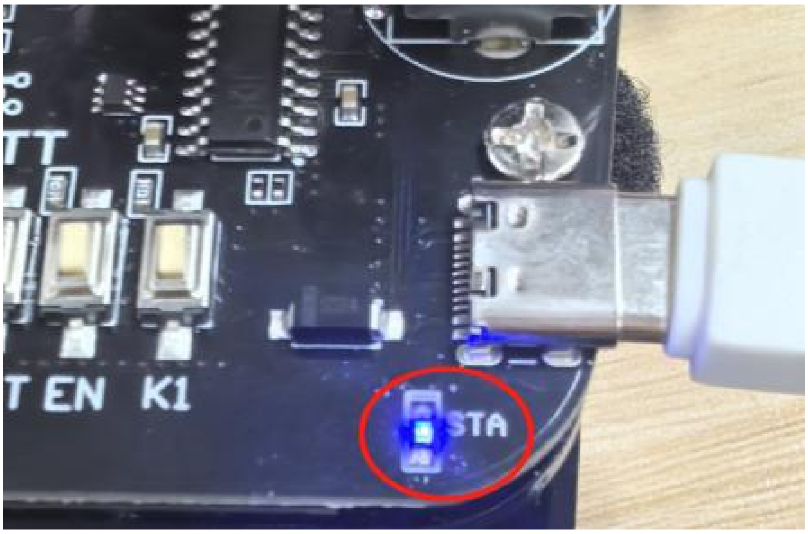
3. Bluetooth Pairing

(1) After uploading the program to the robot, keep the power on.

(2) Make sure the gloves are uploaded Command Sending Program.

(3) Put the motion sensing glove on your right hand, turn on the power of the glove, bring the glove close to the car, and wait for the Bluetooth to automatically connect.

(4) The Bluetooth indicator light will flash during the connection process. If the connection is successful, the Bluetooth indicator light will remain on.



(5) After Bluetooth connection, use the corresponding gestures to control the movement of the car

Lesson 4 Motion-Sensing Glove Control Other Robots

The Motion-Sensing Glove can use the same program to control all ACEBOTT motion robots. The following are the control instructions for the relevant robots.

1. Motion-Sensing Glove Control the Tank Robot Car (QD001+QD004)

The way the Motion-Sensing Glove control QD001+QD004 is the same as the way to control QD001, and the movement of the tracked vehicle is controlled through fixed gestures.

1.1 Control Logic Description

Hand Gesture Pictures	Gesture Description	Gloves Instructions	Car Action
	Open your palms, palms down, lift them up	command11	Go Ahead
	Open your palms, palms down, press down	command12	Back
	Make a fist and rotate your wrist counterclockwise, with your palm facing right.	command13	Rotate Left

	Make a fist and rotate your wrist clockwise, with your palm facing to the left.	command14	Rotate Right
/	/	command0	Stop

1.2 Control Procedures

Click here to open the glove control program for QD001+QD004。

Turn on the power of the crawler car and upload the above program to the crawler car.

1.3 Glove Control

Make sure the gloves are uploaded Command Sending Program, turn on the power of the glove and wait for Bluetooth connection. After Bluetooth is connected, use the gestures in the table above to control the movement of the car.

2. Motion-Sensing Glove Control the Shooting Car (QD001+QD005)

The movement of the car and the shooting movement of the gun are controlled by fixed gestures.

2.1 Control Logic Description

Hand Gesture	Gesture	Gloves	Car Action

Pictures	Description	Instructions	
	Open your palms, palms down, lift them up	command11	Go Ahead
	Open your palms, palms down, press down	command12	Back
	Make a fist and rotate your wrist counterclockwise, with your palm facing right.	command13	Rotate Left
	Make a fist and rotate your wrist clockwise, with your palm facing to the left.	command14	Rotate Right
	Open your hand, palm down, palm tilted to the left	command15	Move Left
	Open your hand, palm down, palm tilted to the right	command16	Move Right
	Extend your index and middle fingers, palm down, and lift up	command5	On the Steering Gear
	Extend your index and middle fingers, palm down, and press down	command6	Under the Steering Gear

	Extend your thumb and index finger, with your palm facing left and up.	command17	Shooting
/	/	command0	Stop

2.2 Control Procedures

Click here to open the glove control program for QD001+QD005.

Turn on the power of the shooting car and upload the above program to the shooting car.

2.3 Glove Control

Make sure the gloves are uploaded Command Sending Program, turn on the power of the glove and wait for Bluetooth connection. After Bluetooth is connected, use the gestures in the table above to control the movement of the shooting car.

3. Motion-Sensing Glove Control the Robotic Arm Car (QD001+QD007)

The movement of the car and the robotic arm is controlled by fixed gestures.

3.1 Control Logic Description

Hand Gesture Pictures	Gesture Description	Gloves Instructions	Car Action

	Extend your index finger, palm down, and lift up	command1	Shoulders Up
	Extend your index finger, palm down, press down	command2	Shoulders Down
	Extend your thumb and index finger, palm down, lean to the right	command3	Chassis Right
	Extend thumb, index finger, palm down, left leaning	command4	Chassis Left
	Extend your index and middle fingers, palm down, and lift up	command5	Elbows Up
	Extend your index and middle fingers, palm down, and press down	command6	Elbow Down
	Extend your index finger, middle finger, ring finger, and pinky finger, palm facing down, leaning left	command7	Claws Open
	Extend your index finger, middle finger, ring finger, and little finger, palm down, and lean to the right	command8	Claw Close

	Extend your index and middle fingers, palm facing right, left	command9	Wrist Left
	Extend your index and middle fingers, with your palm facing left and right	command10	Wrist Right
	Open your palms, palms down, lift them up	command11	Go Ahead
	Open your palms, palms down, press down	command12	Back
	Make a fist and rotate your wrist counterclockwise, with your palm facing right.	command13	Rotate Left
	Make a fist and rotate your wrist clockwise, with your palm facing to the left.	command14	Rotate Right
	Open your hand, palm down, palm tilted to the left	command15	Move Left
	Open your hand, palm down, palm tilted to the right	command16	Move Right
/	/	command0	Stop

3.2 Control Procedures

Click here to open the glove control program for QD001+QD007。

Turn on the power of the solar car and upload the above program to the robotic arm car.

3.3 Glove Control

Make sure the gloves are uploaded Command Sending Program, turn on the power of the glove and wait for Bluetooth connection. After Bluetooth is connected, use the gestures in the table above to control the movement of the robotic arm.

4. Motion-Sensing Glove Control the Solar Car (QD001+QD008)

The movement of the solar car is controlled by fixed gestures.

4.1 Control Logic Description

Hand Gesture Pictures	Gesture Description	Gloves Instructions	Car Action
	Open your palms, palms down, lift them up	command11	Go Ahead
	Open your palms, palms down, press down	command12	Back

	Make a fist and rotate your wrist counterclockwise, with your palm facing right.	command13	Rotate Left
	Make a fist and rotate your wrist clockwise, with your palm facing to the left.	command14	Rotate Right
	Open your hand, palm down, palm tilted to the left	command15	Move Left
	Open your hand, palm down, palm tilted to the right	command16	Move Right
	Extend your index and middle fingers, palm down, and lift up	command5	On the Steering Gear
	Extend your index and middle fingers, palm down, and press down	command6	Under the Steering Gear
/	/	command0	Stop

4.2 Control Procedures

Click here to open the glove control program for QD001+QD008.

Turn on the power of the solar car and upload the above program to the solar car.

4.3 Glove Control

Make sure the gloves are uploaded Command Sending Program, turn on the power of the gloves and wait for Bluetooth connection. After Bluetooth is connected, use the gestures in the table above to control the movement of the solar car.

5. Motion-Sensing Glove Control the Bionic Biped Robot (QD021)

Control the movement of a bipedal robot through fixed gestures.

5.1 Control Logic Description

Hand Gesture Pictures	Gesture Description	Gloves Instructions	Robot Action
	Extend your index finger, palm down, and lift up	command1	Left_Kick
	Extend your index finger, palm down, press down	command2	Right_Kick
	Extend your thumb and index finger, palm down, lean to the right	command3	Right_Tilt
	Extend thumb, index finger, palm down, left leaning	command4	Left_Tilt
	Extend your index and middle fingers, palm down, and lift up	command5	Left_Ankles

	Extend your index and middle fingers, palm down, and press down	command6	Right_Ankles
	Extend your index finger, middle finger, ring finger, and pinky finger, palm facing down, leaning left	command7	Sprint
	Extend your index finger, middle finger, ring finger, and little finger, palm down, and lean to the right	command8	Dance
	Extend your index and middle fingers, palm facing right, left	command9	Left_Stamp
	Extend your index and middle fingers, with your palm facing left and right	command10	Right_Stamp
	Open your palms, palms down, lift them up	command11	Go Ahead
	Open your palms, palms down, press down	command12	Back

	Open your hand, palm down, palm tilted to the left	command15	Turn Left
	Open your hand, palm down, palm tilted to the right	command16	Turn Right
/	/	command0	Stop

5.2 Control Procedures

Click here to open the glove control program for QD021.

Turn on the power of the bipedal robot and upload the above program to the bipedal robot.

5.3 Glove Control

Make sure the gloves are uploaded Command Sending Program, turn on the power of the glove and wait for Bluetooth connection. After Bluetooth is connected, use the gestures in the table above to control the movement of the bipedal robot.

6. Motion-Sensing Glove Control the TruckBott（QE032）

Control the movement of a TruckBott through fixed gestures.

6.1 Control Logic Description

Hand Gesture Pictures	Gesture Description	Gloves Instructions	Robot Action

	Open your palms, palms down, lift them up	command11	Go Ahead
	Open your palms, palms down, press down	command12	Back
	Open your hand, palm down, palm tilted to the left	command15	Turn Left
	Open your hand, palm down, palm tilted to the right	command16	Turn Right
	Extend your index and middle fingers, palm down, and lift up	command5	Forklift Up
	Extend your index and middle fingers, palm down, and press down	command6	Forklift Down
	Extend your index finger, palm down, and lift up	command1	Flip Up
	Extend your index finger, palm down, press down	command2	Flip Down
/	/	command0	Stop

6.2 Control Procedures

Click here to open the glove control program for QE032.

Turn on the power of the TruckBott and upload the above program to the TruckBott

6.3 Glove Control

Make sure the gloves are uploaded Command Sending Program, turn on the power of the glove and wait for Bluetooth connection. After Bluetooth is connected, use the gestures in the table above to control the movement of the TruckBott.

7. Motion-Sensing Glove Control the Quadruped Robot (QD020)

Control the movement of a Quadruped Robot through fixed gestures.

7.1 Control Logic Description

Hand Gesture Pictures	Gesture Description	Gloves Instructions	Robot Action
	Open your palms, palms down, lift them up	command11	Forward
	Open your palms, palms down, press down	command12	Backward

	Open your hand, palm down, palm tilted to the left	command15	Turn Left
	Open your hand, palm down, palm tilted to the right	command16	Turn Right
	Extend your index finger, palm down, and lift up	command1	standby
	Extend your index finger, palm down, press down	command2	lie
	Extend your index and middle fingers, palm down, and lift up	command5	dance2
	Extend your index and middle fingers, palm down, and press down	command6	sleep
	Extend your index finger, middle finger, ring finger, and little finger, palm down, and lean to the right	command8	Hello
	Extend your index and middle fingers, palm facing right, left	command9	dance3

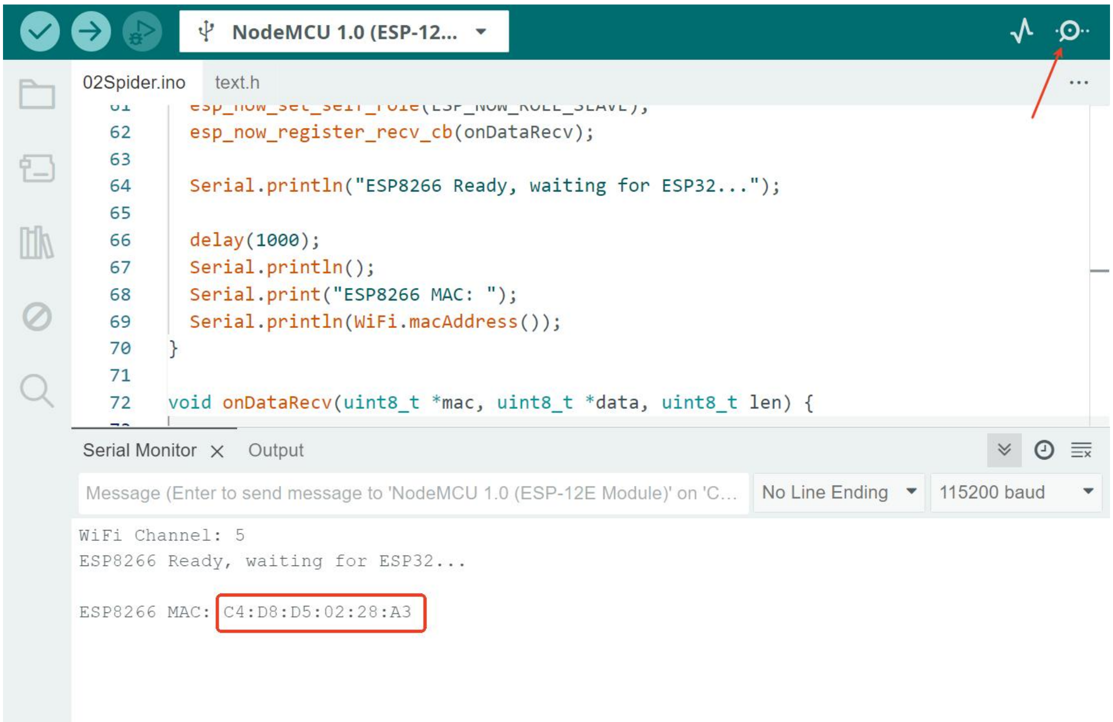
	Extend your index and middle fingers, with your palm facing left and right	command10	dance1
	Make a fist and rotate your wrist counterclockwise, with your palm facing right.	command13	pushup
	Make a fist and rotate your wrist clockwise, with your palm facing to the left.	command14	fighting
/	/	Command0	Stop

7.2 Control Procedures

[Click here to open the glove control program for QD020.]

Turn on the power of the Quadruped Robot and upload the above program to the Quadruped Robot

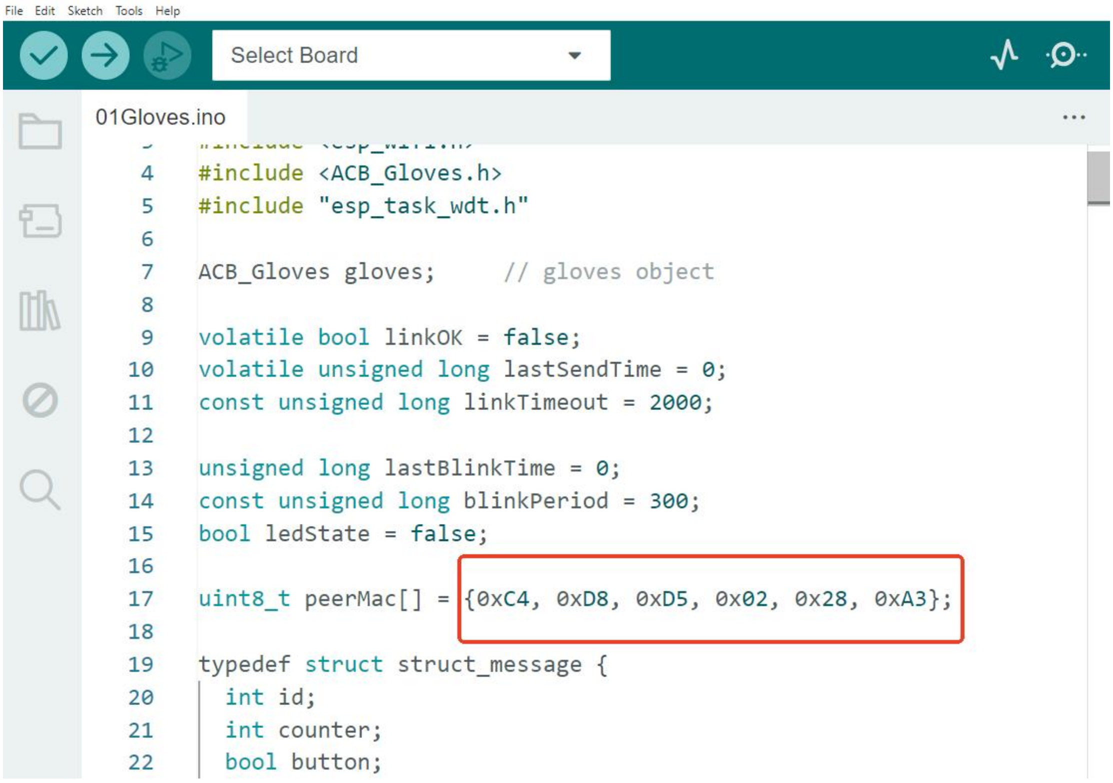
Note: After the program upload is completed, open the serial port monitor. The program will automatically print the MAC address of the ESP8266. If it doesn't print, you can unplug the USB cable and then reconnect it. You need to copy the MAC address and it will be used in the glove program.



7.3 Glove Control

[Click here to open the glove control program for QD023.]

Note: Please fill in the MAC address printed by the spider robot in the following format as specified in the box below, and enter it in the program of the glove. (The "0x" prefix remains unchanged; only the subsequent part needs to be modified.)



Make sure you have changed the MAC address, then upload the program to the glove, turn on the power of the glove, and at this point, use the gestures in the table to control the movement of the quadruped robot.

Note: The blue signal light on the gloves is always on when sending information, and it flashes at other times..

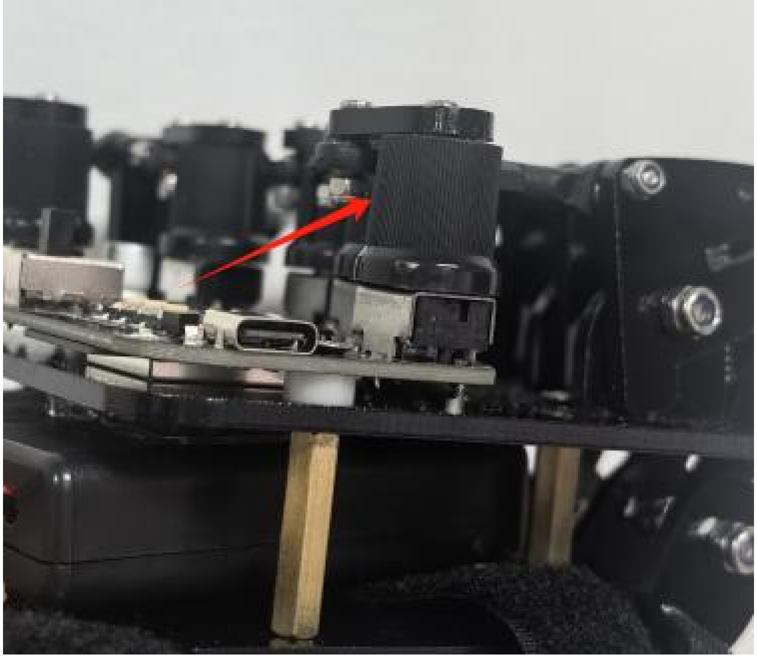
FAQ

Q: What should I do if the motion-sensing glove fails to recognize hand gestures?

A: You can try the following solutions to resolve the issue of unrecognizable hand gestures:

① Adjust the glove's position: Improper wearing may prevent certain gestures from being detected. Try adjusting the glove's tightness or the position of the finger sleeves on the joints.

② Replace the knob: If the screw of the knob is stripped, the glove's detection sensitivity may be reduced. Replace the affected knob with a spare one.



③ Check the battery level: Insufficient battery power can cause detection issues. Observe the Bluetooth indicator light—if it flickers or does not light up, the battery is low. Replace it with a fully charged AAA battery.